Thermochemical Properties of Chemically Reacting Mixtures

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1. Reaction balancing

e.g.

$$2H_2 + O_2 = 2H_2O \tag{R1}$$

Elements must be conserved in a balanced reaction.

For hydrocarbons (C_nH_m), e.g. CH_4, C_2H_4, C_2H_6, C_7H_{16}:

$$C_nH_m + (n + 0.25m)O_2 = nCO_2 + 0.5m\,H_2O \tag{R2}$$

For oxygenated species ($C_iH_jO_k$), e.g. biodiesel

$$C_iH_jO_k + (i + 0.25j - 0.5k)O_2 = iCO_2 + 0.5j\,H_2O \tag{R3}$$

2. Concentration of the ith species in a mixture with K species

Number density (n_i): number of molecules per volume
Mole concentration (C_i): number of moles of the ith species per volume
Mole fraction (X_i):

$$X_i = \frac{C_i}{\sum_{i=1,K} C_i} = \frac{n}{\sum_{i=1,K} n_i}$$

Mass fraction (Y_i):

$$Y_i = \frac{C_i W_i}{\sum_{i=1,K}(C_i W_i)}$$

where W_i is the molecular weight of the ith species.

3. Stoichiometry

A stoichiometric mixture allows complete reaction of both the fuel and the oxidizer in "ideal" cases. The fuel-oxidizer molar ratio of a stoichiometric mixture is denoted as:

$$\left.\frac{[F]}{[O]}\right|_{st}$$

e.g.
for hydrogen in (R1),

$$\left.\frac{[F]}{[O]}\right|_{st} = 2$$

for hydrocarbons in reaction (R2),

$$\left.\frac{[F]}{[O]}\right|_{st} = \frac{1}{\mathrm{n} + 0.25\mathrm{m}}$$

for the oxygenated species in (R3),

$$\left.\frac{[F]}{[O]}\right|_{st} = \frac{1}{\mathrm{i} + 0.25\mathrm{j} - 0.5\mathrm{k}}$$

Equivalence ratio, ϕ: the actual fuel-oxidizer ratio over the stoichiometric fuel-oxidizer ratio

$$\phi = \frac{\dfrac{[F]}{[O]}}{\left.\dfrac{[F]}{[O]}\right|_{st}}$$

Stoichiometric mixture: $\phi = 1$
Lean mixture: $\phi < 1$
Rich mixture: $\phi > 1$

4. Computation of species concentrations for given equivalence ratio

e.g. stoichiometric CH_4 (methane) – air mixture:

$$\frac{C_{CH4}}{C_{O2}} = \frac{[CH_4]}{[O_2]} = 0.5$$

For air:

$$\frac{C_{O2}}{C_{N2}} = \frac{1}{3.76}$$

$$X_{CH4} = \frac{C_{CH4}}{C_{CH4} + C_{O2} + C_{N2}} = \frac{0.5}{0.5 + 1 + 3.76} = \frac{1}{10.52} = 9.5\%$$

$$X_{O2} = 2X_{CH4} = \frac{2}{10.52} = 19.0\%$$

$$X_{N2} = 3.76X_{O2} = \frac{7.52}{10.52} = 71.5\%$$

5. Mixture averaged quantities

Mole averaged quantities $\bar{\alpha}$

$$\bar{\alpha} = \sum_i X_i \alpha_i$$

where α_i is a mole specific quantity.

Mass averaged quantities, $\bar{\beta}$

$$\bar{\beta} = \sum_i Y_i \beta_i$$

where β_i is a mass specific quantity.

Questions:
How to compute mole or mass averaged: internal energy, enthalpy, entropy, c_p, c_v, etc?
How about averaged molecular weight?

6. Other thermodynamic quantities involving multi-component mixtures:

Pressure p, density ρ, temperature T, entropy s, enthalpy h,

Helmholtz free energy (be consistent with molar or mass units!)

$$a = e - Ts$$

Gibbs free energy

$$g = h - Ts$$

partial pressure:

$$p_i = \frac{C_i}{C_{tot}} p = X_i p$$

7. Intensive vs. extensive quantities

Intensive quantifies: property of a mixture that is not dependent on system size, e.g. density, specific volume, pressure, temperature, specific energy/enthalpy/entropy etc

We will try to use lower case symbols for intensive quantities.

Extensive quantities: quantities that is proportional to system size, e.g. volume, total number of molecules, total energy/enthalpy/entropy etc

We will try to use upper case symbols for extensive quantities.

8. The equations relating the thermochemical quantities

Ideal gas law

$$p = \rho RT$$
$$p = nR_uT$$
$$p_i = n_iR_uT$$

where R_u = 8.314 J/mol-K is the universal gas constant, $n = c$ is the species mole concentration.

$$R = \frac{R_u}{\bar{W}}$$

The first law of thermodynamics for a non-reacting mixture

$$dE = dQ - pdV$$

The second law of thermodynamics and the definition of entropy

$$dS = \frac{dQ}{T}$$

For ideal gas:

$$de = c_vdT$$
$$dh = c_pdT$$

BTW:

$$h = e + pv$$
$$c_p = c_v + R$$

Be careful with using consistent mass or molar units of these quantities.

For an ideal gas with constant temperature, i.e. *de* = 0, the specific entropy of a species is:

$$ds = \frac{dQ}{T} = \frac{p_i dv}{T} = -R\frac{dp_i}{p_i}$$

If the partial pressure for the specie change from p_i^0 to p_i,

$$s_i(p_i, T) = s_i^0(p_i^0, T) - Rln\left(\frac{p_i}{p_i^0}\right)$$

Note that when p_i approaches zero, s_i becomes infinity. This is the reason a species can never completely consumed through chemical reactions in an isolated system, because of the second law of thermodynamics.

Using the definition of Gibbs free energy, for constant-temperature varying-pressure mixtures, the free energy is expressed as:

$$g_i(p_i, T) = g_i^0(p_i^0, T) + RTln\left(\frac{p_i}{p_i^0}\right)$$

9. Some misc. terminologies

Isothermal process:

$$T = const$$

Isobaric process:

$$p = const$$

Adiabatic process: no heat exchange between the system and the environment. Note: dQ may not be zero due to chemical heat release.

Isentropic process:

$$dS = 0$$

Based on the definition of entropy, for isentropic process:

$$dQ = 0$$

where dQ is the heat added to the system, either through heat transfer, chemical heat release, or internal friction etc.

isentropic compression: dQ = 0 during the compression.